Toronto Neighbourhood Crime Analysis report

1. Introduction

Background:

Our living social environment is changing over time. By analyzing the Neighbourhood Crime Rates dataset of Toronto city, we gain insights into how crime rates vary across different neighbourhoods and offence categories, and how they have changed over time. We also explore the relationships between different crime types and evaluate the predictive power of past crime patterns for future crime rates. All of this tells us how the overall social environment in Toronto changes, and potentially provides some information for policy making and crime prevention.

Notice: the analysis is intended for descriptive and predictive understanding, not for labeling neighbourhoods or residents

Questions:

- The Main project question:
    - How have Toronto neighbourhood crime patterns changed from 2014 to 2025, and how well can past crime patterns predict future neighbourhood crime rates?

- Supporting subquestions answered during the process:
    1. How much do recorded crime rates vary across Toronto neighbourhoods?
    2. Can neighbourhoods be grouped into meaningful crime-profile clusters?
    3. How well do linear models, random forests, and boosting predict next-year crime rates?

2. Method

Dataset:

This dataset, the Neighbourhood Crime Rates dataset, is from the City of Toronto Open Data Portal. The portal is the City's official open data source (https://www.toronto.ca/city-government/data-research-maps/open-data/), and this dataset contains crime data by neighbourhood. In the data table, each row represents one Toronto neighbourhood, and the columns describe different neighbourhood-level crime measures.

‘This dataset includes the Crime Data by Neighbourhood. Counts are available for Assault, Auto Theft, Break and Enter, Robbery, Theft Over, Homicide and Shooting & Firearm Discharges. Data also includes the crime rate per 100,000 population calculated using the population estimates provided by Environics Analytics.’ quoted from the dataset description page.

Link to the dataset: https://open.toronto.ca/dataset/neighbourhood-crime-rates/

The original data set is obtained from the CKAN API portal (https://ckan0.cf.opendata.inter.prod-toronto.ca) which is in the ‘FOR DEVELOPERS’ sections given by the author.

EDA:

The original dataset has 158 rows and 221 columns. Each row represents a neighbourhood, and the columns include the neighbourhood name, population counts for different years, geometry information of the neighborhood, and crime counts and rates for different offence categories and years.

The dataset has missing values for many columns. A rule is required to cope with them. Table 1 (in the Result section) shows a summary of missing values in the dataset, including “Average missing rate” and “Worst year missing rate” which are the most important conditions for the decision of handling missing values.

Homicide has an extremely high missing rate across all years, with an average missing rate of about 69% and a worst-year missing rate of about 77%. That is too incomplete, so it will be excluded from the main analysis.

Shooting also has a high missing rate, with an average of about 30% and a worst-year missing rate of about 49%. Although not as bad as homicide, it is still quite high and may lead to biased conclusions, so it will not be included in the main analysis either.

For the remaining offences, the missing rate is low. Biketheft has the highest missing rate among remaining offences, but its average missing rate is only about 4.3% and its worst-year missing rate is about 7.0%, which is still relatively small in this dataset. Hence, all of the remaining offences will be included in the main analysis.

That said, even with a low missing rate, these offences with missing values may still have influence on the analysis. Since the dataset has only 158 rows, directly dropping rows with missing values may lead to significant loss of information. Hence, I will handle the missing values based on analysis type. For descriptive analysis, I will omit the missing neighbourhood so I do not ‘make up’ data. For predictive analysis, I will impute the missing values (using median mostly) so the dataset can be used for model training.

Then, I check if the dataset has any impossible values that are considered statistical mistakes, such as negative values in both counts and years and extreme values. The result is that the dataset does not have any, so no decision is required here.

After that, I cleaned all the columns by converting them into lower cases and replacing all space " " with underscore "_" for the later analysis.

Since population affects the counts of offences significantly, analysis will focus on the rate instead of counts to insure unbiasedness.

Moreover, the dataset has multiple dimensions (values of each offence in each year in each neighborhood), so an additional 'melt' dataset which puts everything into a long format is created for later analysis. This long dataset has 13272 rows and 8 columns.

To avoid duplicated code, cleaned/wrangled datasets from the EDA and analysis process are stored in the dataset folders for visualizations to use.

In the last part of the EDA, I inspect the key statistics and distribution of the cleaned dataset. A table (Table 2) and a line chart (Figure 1) are used to show the average crime rate in each year to obtain a first insight of the 'pattern' over time. Then a boxplot (Figure 2) is used to check the distribution focus on the most recent year 2025 to gain an insight only. Having these insights about the dataset helps with latter analysis

Analysis:

The analysis is in two parts, descriptive analysis for the first part of the question, and predictive models to answer the second half.

In the descriptive analysis, K-means clustering was used to group Toronto neighbourhoods into clusters based on similarity of crime features. K-means partitions observations into k clusters by assigning each observation to the nearest cluster centroid and iteratively (recursively) updating those centroids until convergence. The detail algorithm is handled by the skit-learn library.

Before clustering, offence rate variables were rescaled to avoid domination of large values on the clustering solution. This is done because we saw in Figure 2 (Result section) that the data values are highly skewed. Several values of k were explored, and the final number of clusters was selected using the elbow method. In the elbow method, the within-cluster sum of squares (inertia) is plotted against k (Figure 3), and the value of k is then chosen based on the marginal improvement on the model fit (similar to hyperparameter tuning). Because the elbow method does not provide a universal threshold or decision rules, the final choice of k was also adjusted by the model result to avoid extreme imbalance clusters caused by skewed dataset.

Clustering was used to identify neighbourhood crime profiles, but it does not directly measure how crime rates changed over time. Feature engineering is applied here to add an additional feature of slope of changes of each offence in each neighbourhood from 2014 to 2025. Since for specific neighbourhood and offence type, each slope only associates from 12 data points, a simple linear trend was used rather than a more complex time-series model. Specifically, the slope was estimated by fitting a first-degree polynomial to crime rate as a function of year using NumPy's polyfit function.

For the predictive analysis, additional feature engineering was required because the dataset is structured over time and the goal was to use past crime patterns to predict future neighbourhood crime rates. Features we used for predictions includes: for each target offence in year t, the predictors were the rates of all offence types from year t-1 (lag-1) and a year index defined as year minus 2014 (so a scaled time is included in the prediction). These predictors were chosen so the model captures past information about the targeted offence and cross-offence relationships of a neighborhood. The year index was included to allow the models to account for long-term temporal trends.

For model training, validation and testing, the dataset is split into 3 corresponding sets. Since we want the past to predict the future, a traditional proportional random split is not appropriate here. Instead, data of years 2015-2023 is used for model training, 2024 is used for validation and 2025 is used for testing. 2014 data is not used because it does not have 'past' pattern information.

Five models are used to perform the predictions, naive model, linear regression, regression tree, random forest and XGboost. They perform different roles: the naive model is a baseline model which simply assumes no changes at all (predicted value equal to last year value), it provides a simple benchmark against which all other models were compared. Linear regression is constructed as a statistical baseline, it shows if a simple linear relationship is sufficient to explain the future variation. The regression tree provided a basic nonlinear model by recursively splitting the predictor into regions/threshold with similar response values. Random forest is similar , but it extended this idea by constructing many bootstrapped trees, and averaging over them to reduce variance.. XGBoost built trees sequentially so that each new tree focused on correcting the errors of the previous ones. This allows potential more accurate prediction.

Hyperparameter tuning is applied on Random forest and XGBoost on the validation set. It uses a grid of hyperparameters, train models on each combination, the combination of hyperparameters with better performance on the validation set is used for the final model.. Then, Final model performance was evaluated on the 2025 test set using RMSE, MAE, and R^2.

This predictive framework was applied separately to each offence category (there are 7 of them after cleaning). The hyperparameter tuning is also applied separately to each offence type to avoid biasness.

3. Result

Result from EDA:

	Offence	Measure Type	Years Available	Average Missing Count	Worst-Year Missing Count	Worst Year	Average Missing Rate (%)	Worst-Year Missing Rate (%)
0	Homicide	Count	12	108.67	121	2025	68.78	76.58
1	Homicide	Rate	12	108.67	121	2025	68.78	76.58
2	Shooting	Count	12	48.08	78	2014	30.43	49.37
3	Shooting	Rate	12	48.08	78	2014	30.43	49.37
4	Biketheft	Count	12	6.83	11	2025	4.32	6.96
5	Biketheft	Rate	12	6.83	11	2025	4.32	6.96
6	Theftover	Count	12	3.25	9	2016	2.06	5.70
7	Theftover	Rate	12	3.25	9	2016	2.06	5.70
8	Robbery	Count	12	0.58	2	2022	0.37	1.27
9	Robbery	Rate	12	0.58	2	2022	0.37	1.27
10	Autotheft	Count	12	0.08	1	2017	0.05	0.63
11	Autotheft	Rate	12	0.08	1	2017	0.05	0.63
12	Assault	Count	12	0.00	0	2014	0.00	0.00
13	Assault	Rate	12	0.00	0	2014	0.00	0.00
14	Breakenter	Count	12	0.00	0	2014	0.00	0.00
15	Breakenter	Rate	12	0.00	0	2014	0.00	0.00
16	Population	Count	1	0.00	0	2025	0.00	0.00
17	Theftfrommv	Count	12	0.00	0	2014	0.00	0.00
18	Theftfrommv	Rate	12	0.00	0	2014	0.00	0.00

Table 1: Summary of missing values of the dataset

	mean_rate						
offence	**ASSAULT**	**AUTOTHEFT**	**BIKETHEFT**	**BREAKENTER**	**ROBBERY**	**THEFTFROMMV**	**THEFTOVER**
year							
2014	610.45	125.78	132.68	268.16	137.85	353.40	37.41
2015	657.35	119.86	134.33	255.72	125.32	319.87	37.06
2016	657.29	116.49	148.27	229.90	129.64	271.30	38.11
2017	660.62	124.88	149.36	243.43	138.81	288.04	40.29
2018	666.58	161.10	152.53	260.23	126.40	306.73	42.15
2019	695.52	174.60	136.77	289.70	118.86	335.39	46.31
2020	597.54	190.38	148.37	239.45	91.37	343.76	40.73
2021	630.53	220.39	117.76	191.50	75.05	269.57	35.95
2022	685.60	319.48	110.32	202.15	91.90	300.51	46.99
2023	765.88	385.88	99.71	249.20	99.16	267.82	53.01
2024	778.45	285.41	90.98	216.79	97.44	233.76	54.34
2025	745.50	216.51	69.11	189.86	76.40	207.86	56.83

Table 2: Offence rate in each year averaged over all neighborhoods.

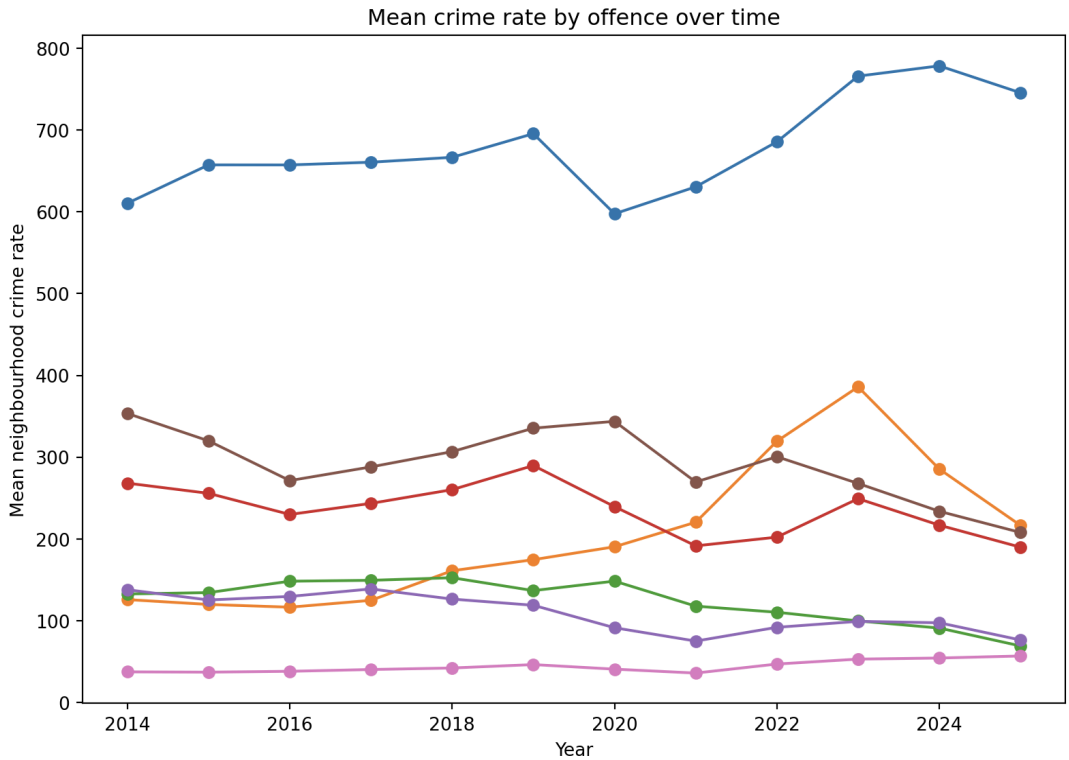

Figure 1: Line chart showing the information from Table 2.

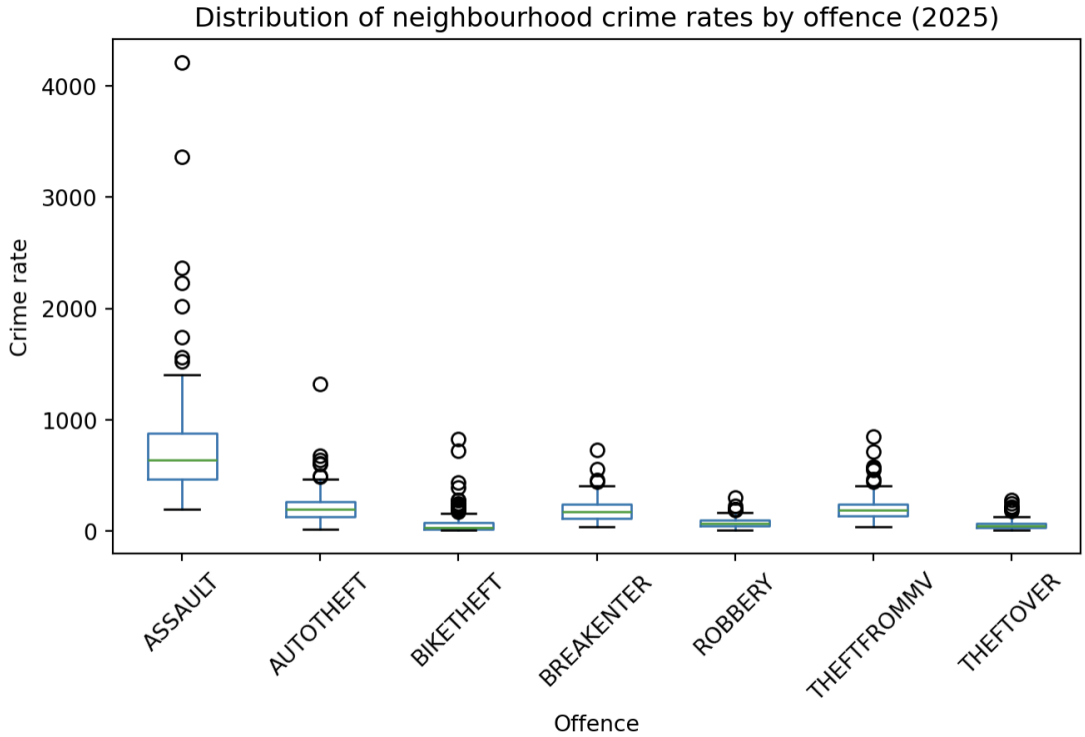

Figure 2: boxplot that shows distribution of crime rate in 2025

Result from descriptive analysis:

Figure 3 shows the result from the elbow method. As shown, the improvement continues to increase with lower extent. Moderate reduction in variance from k=4 to k=6, and low reduction from k=6 to k= 8. Therefore, k=4 and k=6 are both tried. The k=4 is finally chosen because 6 cluster will make some clusters to have only 1 or 2 neighbourhoods while some have handraids.

The result of the model is shown in Table 3. In summary:

- **Cluster 0 - low overall crime cluster.** It has a relatively low overall crime rate compared to other clusters.
- **Cluster 1 - high property-crime clusters.** It has especially high rates of autotheft, theftfrommv, and theftover.
- **Cluster 2 - extreme biketheft and assault cluster.** It stands out sharply from all other clusters because it has by far the highest assault rate and an extremely high biketheft rate.
- **Cluster 3 - moderately elevated mixed-crime cluster.** It has higher rates than Cluster 0 across most offence types,

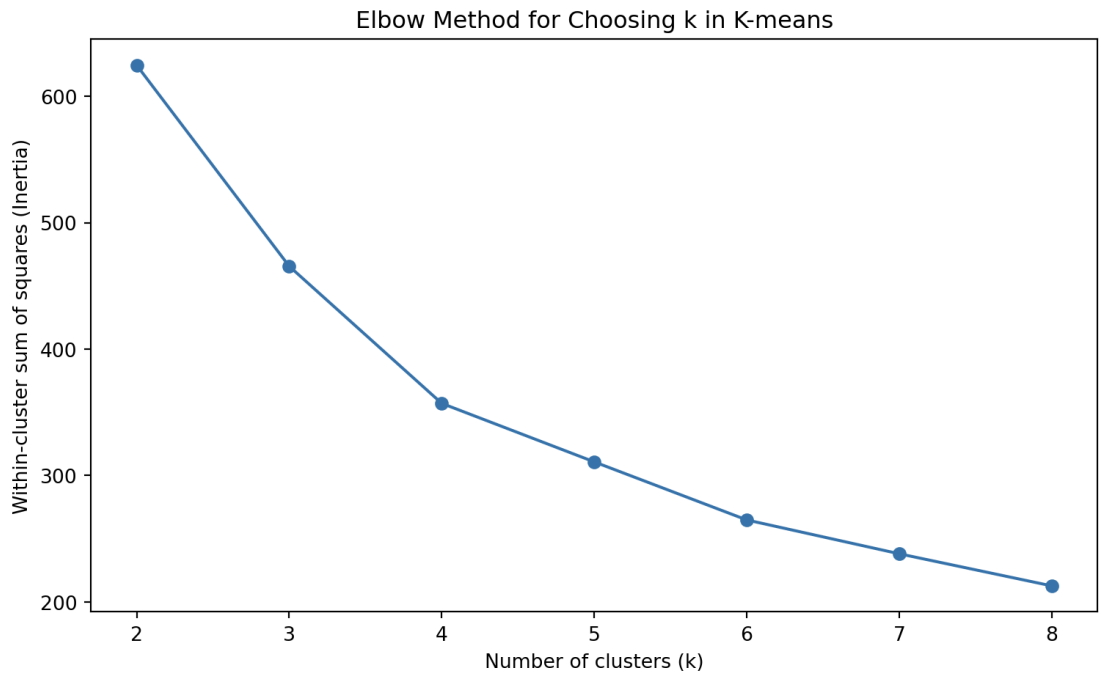

Figure 3: elbow method result

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Cluster profiles (mean crime rates on original scale):
offence  ASSAULT  AUTOTHEFT  BIKETHEFT  BREAKENTER  ROBBERY  THEFTFROMMV  \
cluster
0          490.7      193.1       57.7       186.9     80.7        233.0
1          936.8      759.2       60.3       379.4    196.6        620.4
2         2533.7      168.8      921.0       702.2    412.7        936.5
3          972.6      163.2      211.1       305.6    141.7        338.3

offence  THEFTOVER
cluster
0             31.9
1            161.5
2            155.1
3             49.1
```

Table 3: K-mean Clustering result

However, this result is not clear and intuitive enough. Since the dataset contains the geometry information, an interactive geomap showing the result is constructed on the website. Colors are used to distinguish each cluster.

Link to the Cluster map on the website:

https://jerry-zeng-uoft.github.io/JSC370_Project/cluster-map.html

The cluster map suggests that most of the neighbourhood is in the low crime rate Cluster 0, except the downtown area which is mostly Cluster 3. Only a few west downtown regions are in Cluster 2 with high biketheft and assault, and only a few west Toronto regions are in Cluster 1 with high property-crime.

The slope (changes from 2014 to 2025) is also displayed using the real geometric map. A diverging color scale is used to illustrate the slope, where blue represents decreasing crime rate over time, red/pink represents increasing crime rates, and grey/natural light represents near zero slope.

Link to the Slope map on the website:

https://jerry-zeng-uoft.github.io/JSC370_Project/slope-map.html

The slope map suggests that the crime rate pattern changes a lot from 2014 to 2015. Overall, Assault, Autotheft and theftover rate were increasing among most of the neighborhoods. Biketheft and Theftoverfrommv rates have decreased over past years. Breakover and Robbery are more complex, with some neighbourhoods having increasing rate and some having decreasing rates.

Result from predictive analysis:

The hyperparameter tuning is applied separately to each offence type to avoid biasness caused by using one combination for all categories. There will be 7 x 36 combinations in total. Hence, only the result of the tuning will be shown here in Table 4 and Table 5 for Random Forest and XGboost respectively.

offence	model	n_train_val	n_test	n_estimators	max_depth	min_samples_leaf	max_features	test_rmse	test_mae	test_r2	
0	assault	Random forest (tuned)	1478	149	200	8	3	0.5	205.771	114.345	0.842
1	autotheft	Random forest (tuned)	1477	149	500	4	1	sqrt	100.856	86.794	0.564

2	bikethefft	Random forest (tuned)	1432	140	200	8	3	0.5	52.708	31.998	0.777
3	breakenter	Random forest (tuned)	1478	149	200	4	5	0.5	68.017	51.463	0.564
4	robbery	Random forest (tuned)	1473	149	200	8	5	0.5	49.232	36.371	-0.099
5	theftfrommv	Random forest (tuned)	1478	149	500	4	3	0.5	105.598	82.760	0.218
6	theftover	Random forest (tuned)	1449	149	500	8	3	sqrt	25.109	18.468	0.675

Table 4: result from hyperparameter tuning of random forest.

offence	colsample_bytree	learning_rate	max_depth	n_estimators	subsample	val_rmse	val_mae	val_r2	
0	assault	1.0	0.10	4	100	1.0	196.637832	122.451314	0.849268
1	autotheft	0.8	0.03	6	100	1.0	193.647291	142.685166	-0.010485
2	biketheft	1.0	0.10	4	100	1.0	47.922889	31.006218	0.892168
3	breakenter	1.0	0.03	2	100	1.0	76.058303	58.476166	0.596536
4	robbery	1.0	0.03	4	100	1.0	45.286955	37.215404	0.615536
5	theftfrommv	0.8	0.10	2	100	0.8	111.062939	74.262539	0.372509
6	theftover	1.0	0.03	2	100	1.0	28.428496	19.555024	0.616351

Table 5: result from hyperparameter tuning of XGboost.

The interactive histogram in the link shows all the result metrics for all 5 models and 7 offence categories.

Link to the Predictive model comparison on the website:

https://jerry-zeng-uoft.github.io/JSC370_Project/model-comparison.html

The model comparison suggests that predictive performance varies significantly across offence types. For **Assault**, all five models perform very well, with R^2 values around 0.84–0.92 and relatively high but still acceptable RMSE/MAE values. This indicates that past crime patterns are highly informative in the prediction of future assault rates. For **Biketheft**, the models also perform fairly well, especially linear regression, which has the lowest error and the highest R^2, suggesting that a simple linear relationship is already effective for this type of offence. **Breakenter** and **Theftover** perform moderately: all models have similar R^2 values, and linear regression appears to perform best or near best, implying that these two offences are less predictable than assault or biketheft using past pattern information. **Theftfrommv** is more difficult to predict with models we have, with generally lower R^2 values. Among all models, XGBoost performs best for this offence. This suggests that a more complex tree structure may matter for this offence. **Autotheft** and **Robbery are the most problematic offences in prediction**. For autotheft, although the naive baseline, random forest, and especially XGBoost already have bad performance, they are much better than linear regression and the regression tree. Linear regression and the single tree even generates negative R^2, meaning they are even worse than just predicting with the mean. For robbery, performance is weak for all models, with negative R^2 for most models except linear regression, which is only 'near' positive. This implies that the past provides limited predictive power for future robbery rates.

Overall, the comparison shows that no single model is uniformly best across all offences. XGBoost appears to be the strongest overall model especially in autotheft, assault, and theftfrommv. Linear regression performs surprisingly well for biketheft, breakenter, robbery, and theftover, showing that simpler models can still be highly competitive. Random forest improves on some offences but is not well performed overall, while the single regression tree is generally weaker than the other two trees. The naive baseline is surprising. It is a good model for some offences, especially assault, which means that year-to-year persistence alone explains a lot of variation already. Taken together, these results suggest that past crime patterns can predict some future offence rates reasonably well, but predictive power differs vary much based on the offence type.

Moreover, the feature importance bar chart gives additional information of the comparison of importance of each predictor.

Link to the Feature importance on the website:

https://jerry-zeng-uoft.github.io/JSC370_Project/feature-importance.html

The importance chart suggests that all categories except breakenter have the lag-1 rate of itself as the most important feature for both Random forest and XG boost. The only exception is the breakenter with XGboost, with lag-1 assault slightly more important than lag-1 breakenter. For most of the offence types, the past pattern of other types and even the time index are considered not important in the prediction.

4. Conclusion and Summary

In conclusion, How have Toronto neighbourhood crime patterns changed from 2014 to 2025? From the analysis, there was indeed a change in the pattern, but probability not on the overall rate. While the overall crime rate was stable, some types of crime rate rose while some dropped. Also, different neighbourhoods experienced different changes in the crime patterns.

How well can past crime patterns predict future neighbourhood crime rates? The predictions differ very much depending on the offence types. Different models have slight distinction in performance, but the offence type itself still affects the prediction the most. For robbery, Theftfrommv and Autotheft, the past crime pattern shows low predictive power to the future crime rate, whereas for Assault, Biketheft, Breakenter and Theftover, most models show strong prediction accuracy.

The biggest limitation is that the dataset contains only the crime data itself, without other neighbourhood information such as income, demographics, housing conditions, transit access and policing-related variables. All these could be very powerful in predicting the future crime rate, and the change of them over time could be a good point to analyze associations. Because of this, the predictive models rely mainly on past crime patterns.

A second limitation is the relatively short time dimension. The dataset covers 2014 to 2025, and gives only 12 yearly observations per neighbourhood-offence series. After feature engineering and train test split, this is even worse. This is enough for simple trend summaries, but it limits the reliability of more complex temporal models.

Another limitation is missingness. Homicide and shooting had to be excluded because of high missing rates, and even the retained offences contain some incomplete observations. Although these were handled carefully, missing data may still reduce comparability across neighbourhoods and weaken some analyses.